MAT3003 PSR Module 3

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February 15, 2025

1 Discrete Bivariate RV

Question 1: For the following bivariate probability distribution of X and Y, calculate the required probabilities.

$X \backslash Y$	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Find:

1. $\mathbb{P}(X \leq 1, Y = 2)$
2. $\mathbb{P}(X \leq 1)$
3. $\mathbb{P}(Y = 3)$
4. $\mathbb{P}(Y \leq 3)$
5. $\mathbb{P}(X < 3, Y \leq 4)$

Question 2: Given the following bivariate probability distribution:

$Y \backslash X$	-1	0	1
0	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
1	$\frac{3}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
2	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$

Find:

1. The marginal distributions of X and Y.
2. The conditional distribution of X given $Y = 2$.

Question 3: Given the joint density function:

$$f(x, y) = \frac{(2x + y)}{127}$$

where X and Y take integer values $0, 1$, and 2. Find the conditional distribution of Y given $X = x$.

Question 4: Two ideal dice are thrown. Let X_1 be the score on the first die and X_2 be the score on the second die. Let Y denote the maximum of X_1 and X_2, i.e.,

$$Y = \max(X_1, X_2)$$

Find:

1. The joint distribution of Y and X_1.
2. The mean and variance of Y and the covariance $\mathrm{Cov}(Y, X_1)$.

2 Continuous Bivariate RV

Question 5: Let the joint probability density function (pdf) of X and Y be:

$$f(x, y) = \frac{3}{16}xy^2, \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 2.$$

Find:

1. The marginal probability density functions $f_X(x)$ and $f_Y(y)$.
2. Are X and Y independent? Justify your answer.
3. The means and variances of X and Y.
4. The probability $P(X \leq Y)$.

Question 6: Let X and Y have the joint pdf:

$$f(x, y) = x + y, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

Find:

1. The marginal pdfs $f_X(x)$ and $f_Y(y)$ and show that $f(x, y) \neq f_X(x)f_Y(y)$. Thus, X and Y are dependent.
2. Compute:
   - μ_X, the mean of X.
   - μ_Y, the mean of Y.
   - σ_X^2, the variance of X.
   - σ_Y^2, the variance of Y.

Question 7: Let the joint pdf of X and Y be:

$$f(x,y) = 2e^{-x-y}, \quad 0 \leq x \leq y < \infty.$$

Find:

1. The marginal pdfs $f_X(x)$ and $f_Y(y)$.
2. Are X and Y independent?

Question 8: Let the joint pdf of X and Y be:

$$f(x,y) = \frac{3}{2}, \quad x^2 \leq y \leq 1, \quad 0 \leq x \leq 1.$$

Find:

1. $P(0 \leq X \leq 1/2)$.
2. $P(1/2 \leq Y \leq 1)$.
3. $P(X \geq 1/2, Y \geq 1/2)$.
4. Are X and Y independent? Justify your answer.

3 Regression Analysis

Question 9: Calculate the regression coefficients and obtain the regression lines for the following data:

X	1	2	3	4	5	6	7
Y	9	8	10	12	11	13	14

Hence, find the value of Y when $X = 2.5$ and the value of X when $Y = 12.5$.

Question 10: In a laboratory experiment on correlation research, the equation of the two regression lines were found to be:

$$2X - Y + 1 = 0 \quad \text{and} \quad 3X - 2Y + 7 = 0.$$

Find:

- The means of X and Y.
- The regression coefficients.
- The correlation coefficient between the two variables X and Y.

Question 11: Given the regression equations:

$$Y = \frac{3}{2}X + \frac{7}{2}, \quad X = (1+c)Y - \frac{1}{2}.$$

Find the range of c.

Question 12: The equations of two regression lines are:

$$3X + 2Y - 26 = 0, \quad 6X + Y - 31 = 0.$$

Find:

1. The mean of X and Y.
2. The correlation coefficient between X and Y.
3. Estimate of Y for $X = 2$.
4. The variance of X if the variance of Y is 36.